

Software Requirements Specification

Flight Reservations System

Mapping Flights, Airports, Passenger Seating and Ticket Classes

Document	Software Requirements Specification (SRS)
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1. Introduction

1.1 Purpose

This document specifies the requirements for a Flight Reservations System: a web application that lets passengers search for flights between airports, choose a ticket class, select seats, and book tickets, and lets administrators manage airports, flights, aircraft and pricing. It is the reference for design, development and testing.

1.2 Scope

The system will map four core entities and the relationships between them: **Airports, Flights, Passenger Seating** and **Ticket Classes**. Included: search, seat selection, booking, e-ticket generation, cancellation, and admin management. Simulated for this project: payment processing. Out of scope for v1.0: loyalty programs, hotel and car rental, real airline (GDS) integrations, and mobile native apps.

1.3 Definitions

Term	Meaning
IATA code	3-letter airport code (e.g. JFK, LHR)
PNR	Passenger Name Record; the unique booking reference (6 characters)
Ticket class	Fare category: Economy, Premium Economy, Business, First
Seat map	Layout of all seats on an aircraft, by row and column
Flight instance	One scheduled operation of a route on a specific date and time

1.4 Intended Readers

Developer (implementation), testers, and portfolio reviewers who want to understand the design decisions.

2. Overall Description

2.1 Product Perspective

A standalone web application with a browser front end, a back-end API, and a relational database. It is a clone-style product modeled on common airline booking sites.

2.2 User Classes

User class	Description	Key needs
Guest	Unregistered visitor	Search flights and view fares
Passenger (registered)	Logged-in customer	Book, pick seats, view/cancel bookings
Administrator	Staff member	Manage airports, flights, aircraft, seat maps, prices

2.3 Operating Environment

- Modern browsers (Chrome, Firefox, Safari, Edge; latest two versions), responsive from 360 px wide.
- Server-side application with a relational database (e.g. PostgreSQL or SQLite for development).

2.4 Assumptions and Constraints

- All flight times are stored in UTC and displayed in the local time of each airport.
- Payments are simulated (no real card data is stored).
- A seat can belong to only one active booking per flight instance.
- Each aircraft has exactly one seat-map configuration.

3. System Data Model (Mapping)

This section defines how airports, flights, seats and ticket classes map to each other. Each entity below becomes a database table.

3.1 Entities

Entity	Key attributes
Airport	airport_id (PK), iata_code (unique), name, city, country, timezone, latitude, longitude
Aircraft	aircraft_id (PK), model, registration, total_seats, seat_map_id (FK)
Route	route_id (PK), origin_airport_id (FK), destination_airport_id (FK), distance_km, duration_min
Flight	flight_id (PK), flight_number, route_id (FK), aircraft_id (FK), departure_utc, arrival_utc, status
TicketClass	class_id (PK), name (Economy / Premium Economy / Business / First), baggage_allowance_kg, refundable, changeable, perks
FlightFare	fare_id (PK), flight_id (FK), class_id (FK), base_price, seats_available
Seat	seat_id (PK), aircraft_id (FK), row_number, column_letter, class_id (FK), seat_type (window / aisle / middle), is_exit_row, extra_legroom
Passenger	passenger_id (PK), first_name, last_name, date_of_birth, passport_number, email, phone
User	user_id (PK), email (unique), password_hash, role (guest / passenger / admin)
Booking	booking_id (PK), pnr (unique), user_id (FK), status, total_amount, created_at
Ticket	ticket_id (PK), booking_id (FK), passenger_id (FK), flight_id (FK), class_id (FK), seat_id (FK), price_paid, ticket_number, status
Payment	payment_id (PK), booking_id (FK), amount, method, status, paid_at

3.2 Relationships

Relationship	Cardinality	Notes
Airport to Route	1 : many (as origin and as destination)	A route must connect two different airports
Route to Flight	1 : many	Same route runs on many dates
Aircraft to Flight	1 : many	An aircraft cannot be on two overlapping flights
Aircraft to Seat	1 : many	Seat map is defined per aircraft
TicketClass to Seat	1 : many	Each seat belongs to one class (e.g. rows 1-4 Business)
Flight to TicketClass (via FlightFare)	many : many	Each flight has a price and availability per class
Booking to Ticket	1 : many	One booking can hold several passengers

Relationship	Cardinality	Notes
Ticket to Seat and Flight	many : 1	Unique on (flight_id, seat_id) for active tickets

3.3 Ticket Classes

Class	Typical seat layout	Baggage	Refund	Change	Perks
Economy	3-3 (or 3-4-3)	1 x 23 kg	No / fee	Fee	Standard meal
Premium Economy	2-4-2	2 x 23 kg	Partial	Reduced fee	Extra legroom, priority boarding
Business	2-2-2 or 1-2-1	2 x 32 kg	Yes	Free	Lounge, lie-flat seat, premium meal
First	1-1-1	3 x 32 kg	Yes	Free	Suite, lounge, chauffeur (optional)

3.4 Passenger Seating Map

Seats are identified by row number and column letter (e.g. 12A). The seat map stores, for each seat: class, type (window / aisle / middle), exit-row flag and availability per flight. Seat status on a specific flight is one of: **Available**, **Held** (temporarily locked during checkout, for 10 minutes), **Booked**, **Blocked** (crew or maintenance).

3.5 Class Diagram

Conceptual class model of the core booking entities (User, Booking, Flight, Passenger, Payment) and their multiplicities. It is a simplified view of the full data model in Section 3.1.

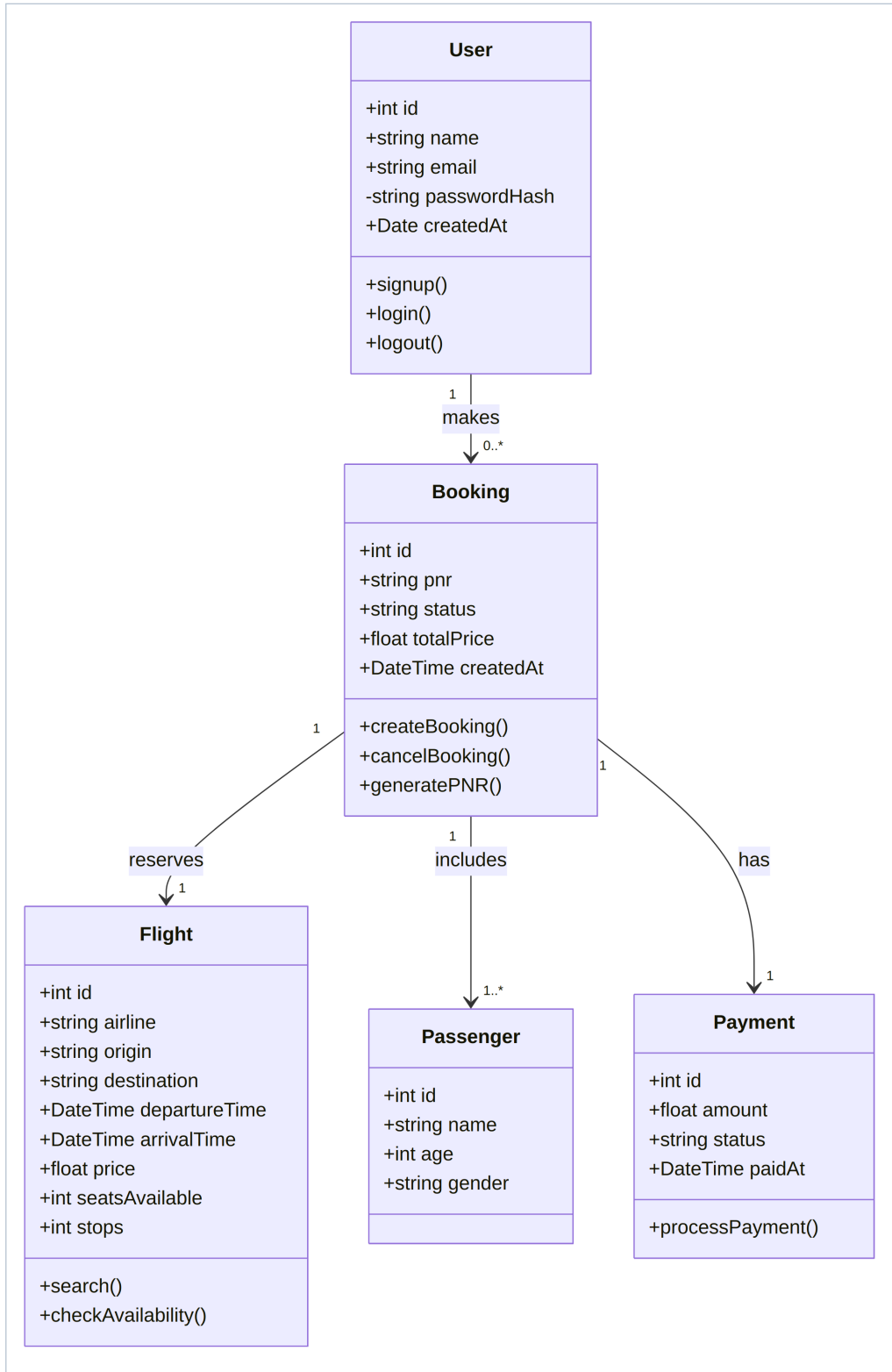


Figure 3.1 - Class diagram

3.6 Booking Status Lifecycle

State diagram showing how a booking moves from Draft to Completed or Cancelled, including the payment retry loop.

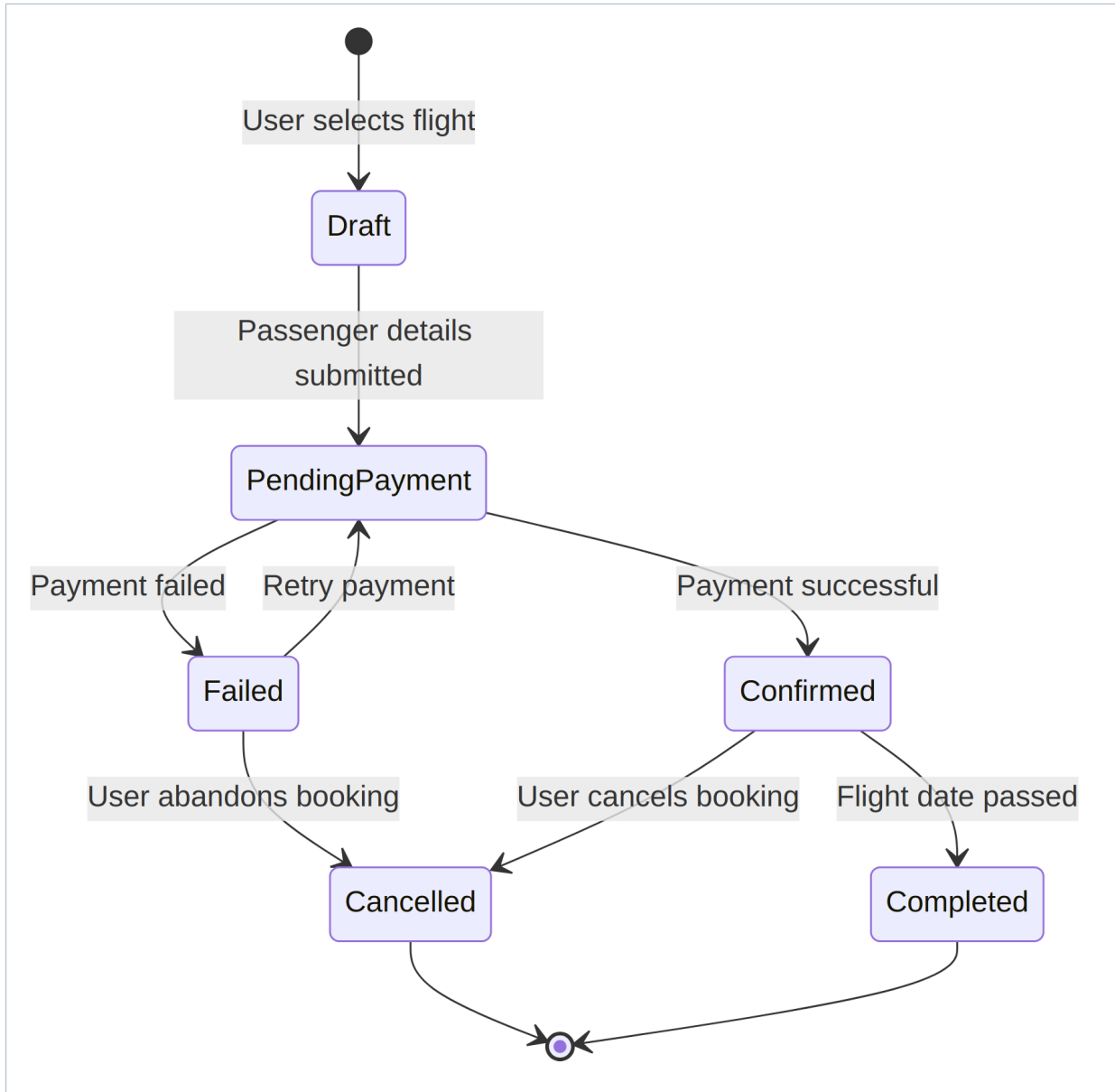


Figure 3.2 - Booking state diagram

4. Functional Requirements

Priority: **M** = Must, **S** = Should, **C** = Could.

4.1 Account and Access

ID	Requirement	Priority
FR-1.1	The system shall allow users to register with email and password.	M
FR-1.2	The system shall allow registered users to log in and log out securely.	M
FR-1.3	The system shall support roles: guest, passenger, administrator.	M
FR-1.4	The system shall allow password reset by email.	S

4.2 Airport Management

ID	Requirement	Priority
FR-2.1	Administrators shall create, edit and deactivate airports with IATA code, name, city, country and timezone.	M
FR-2.2	The system shall reject duplicate IATA codes.	M
FR-2.3	Users shall search airports by code, city or name with auto-complete.	M
FR-2.4	The system shall display airports on a map using coordinates.	C

4.3 Flight and Route Management

ID	Requirement	Priority
FR-3.1	Administrators shall create routes between two different airports.	M
FR-3.2	Administrators shall schedule flights with flight number, route, aircraft, departure and arrival times.	M
FR-3.3	The system shall prevent assigning one aircraft to overlapping flights.	M
FR-3.4	Administrators shall set flight status: Scheduled, Delayed, Cancelled, Departed, Arrived.	M
FR-3.5	Administrators shall support recurring schedules (e.g. daily, weekly).	S

4.4 Flight Search

ID	Requirement	Priority
FR-4.1	Users shall search by origin, destination, departure date and number of passengers.	M
FR-4.2	The system shall support one-way and round-trip searches.	M
FR-4.3	Results shall show flight number, times, duration, price and remaining seats per class.	M
FR-4.4	Users shall sort results by price, duration and departure time, and filter by ticket class.	S
FR-4.5	The system shall show connecting itineraries with one stop.	C

4.5 Ticket Classes and Fares

ID	Requirement	Priority
FR-5.1	Administrators shall define ticket classes and their rules (baggage, refund, change).	M
FR-5.2	Administrators shall set a price and seat quota per class for each flight.	M
FR-5.3	The system shall show a class comparison table during booking.	S
FR-5.4	The system shall reduce seats_available whenever a ticket is confirmed and restore it on cancellation.	M

4.6 Seat Map and Seat Selection

ID	Requirement	Priority
FR-6.1	Administrators shall define an aircraft seat map with rows, columns, class zones and exit rows.	M
FR-6.2	The system shall display a visual seat map per flight showing Available, Held, Booked and Blocked seats.	M
FR-6.3	A passenger shall only select seats belonging to the ticket class purchased.	M
FR-6.4	The system shall hold a selected seat for 10 minutes during checkout and release it on timeout.	M
FR-6.5	The system shall prevent double-booking of any seat on the same flight.	M
FR-6.6	The system shall enforce exit-row rules (e.g. minimum age 15; no infants).	S
FR-6.7	The system shall auto-assign seats if the passenger skips selection.	S
FR-6.8	Users shall change their seat before check-in closes.	S

4.7 Booking and Payment

ID	Requirement	Priority
FR-7.1	Users shall enter passenger details (name as on passport, date of birth, passport number, contact).	M
FR-7.2	The system shall show a booking summary with a price breakdown (base fare, taxes, fees).	M
FR-7.3	The system shall process a simulated payment and mark the booking Confirmed or Failed.	M
FR-7.4	The system shall generate a unique PNR and ticket numbers per passenger.	M
FR-7.5	The system shall send a confirmation email with an e-ticket (PDF).	S
FR-7.6	The system shall support up to 9 passengers per booking.	M

4.8 Booking Management

ID	Requirement	Priority
FR-8.1	Passengers shall view their upcoming and past bookings.	M
FR-8.2	Passengers shall retrieve a booking by PNR and last name.	M
FR-8.3	Passengers shall cancel a booking; refunds follow the ticket class rules.	M
FR-8.4	The system shall notify passengers when a flight is delayed or cancelled.	S
FR-8.5	Passengers shall check in online 24 hours to 1 hour before departure and get a boarding pass.	C

4.9 Administration and Reports

ID	Requirement	Priority
FR-9.1	Administrators shall view all bookings and passenger manifests per flight.	M
FR-9.2	The system shall report load factor (seats sold / total seats) and revenue per flight and class.	S

5. External Interface Requirements

5.1 User Interface

- Home page with a search form (from, to, dates, passengers, class).
- Results page with sortable flight cards.
- Seat-selection page with a graphical, color-coded seat map and legend.
- Passenger details, review and payment, confirmation, and My Bookings pages.
- Admin dashboard for airports, aircraft/seat maps, flights and fares.
- The UI shall meet WCAG 2.1 AA guidelines (keyboard navigation, contrast, labels).

5.2 Software Interfaces

- REST API (JSON) between front end and back end.
- Email service (SMTP or provider API) for confirmations.
- Optional: mock payment gateway; optional maps library for airport display.

5.3 Key API Endpoints (draft)

Method and path	Purpose
GET /airports?q=	Search airports
GET /flights?from=&to;=&date;=&class;=	Search flights
GET /flights/{id}/seats	Seat map and seat status for a flight
POST /bookings	Create a booking with passengers and seats
POST /bookings/{id}/pay	Simulated payment
GET /bookings/{pnr}	Retrieve a booking
DELETE /bookings/{id}	Cancel a booking
POST /admin/flights	Create a flight (admin)

6. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	Search results shall return within 3 seconds for typical data; seat map shall load within 2 seconds.
NFR-2	Concurrency	Seat booking shall be atomic so two users cannot book the same seat (database unique constraint and transactions).
NFR-3	Security	Passwords shall be hashed (bcrypt or argon2); all traffic over HTTPS; protection against SQL injection, XSS and CSRF.
NFR-4	Privacy	Passport and personal data shall be visible only to the owner and administrators; no real card data stored.
NFR-5	Availability	Target 99% uptime for the deployed demo.
NFR-6	Usability	A first-time user shall complete a booking in under 5 minutes; responsive layout for mobile.
NFR-7	Maintainability	Modular code, README, and automated tests for core booking and seating logic (aim 70%+ coverage).
NFR-8	Data integrity	Foreign keys and constraints enforce valid relationships; all times stored in UTC.
NFR-9	Portability	Deployable with one command (e.g. Docker) or a documented setup.

7. Business Rules

ID	Rule
BR-1	Origin and destination airports must differ.
BR-2	A flight's arrival time must be after its departure time.
BR-3	Booking closes 1 hour before departure.
BR-4	Infants (under 2) travel on an adult's lap and do not occupy a seat; there is at most one infant per adult.
BR-5	Unaccompanied minors under 12 are not permitted in v1.0.
BR-6	Held seats expire after 10 minutes without payment.
BR-7	Cancellation refunds: Economy 0-50%, Premium 50%, Business/First 100% (configurable per class).
BR-8	Total price = base fare + taxes and fees + optional seat surcharge.

8. Use Cases

UC-1: Book a Flight

Item	Detail
Actor	Passenger
Precondition	Passenger is logged in; flights exist for the requested route and date
Main flow	1. Enter origin, destination, date and passengers. 2. View results and pick a flight. 3. Choose ticket class. 4. Enter passenger details. 5. Select seats on the seat map. 6. Review price. 7. Pay. 8. Receive PNR and e-ticket.
Alternate flows	A1: Class sold out, so the system suggests another class or date. A2: Seat already taken, so the system asks for another seat. A3: Hold expires, so seats are released and the user restarts seat selection. A4: Payment fails, so the booking is not confirmed and seats are released.
Postcondition	Booking is Confirmed; seats are Booked; availability is updated

UC-2: Cancel a Booking

Item	Detail
Actor	Passenger
Main flow	1. Open My Bookings. 2. Select booking. 3. Choose Cancel. 4. System shows refund amount by class rules. 5. Confirm. 6. Seats released, refund recorded, email sent.
Postcondition	Booking is Cancelled; seats return to Available

UC-3: Schedule a Flight

Item	Detail
Actor	Administrator
Main flow	1. Select a route (or create airports and a route). 2. Choose aircraft. 3. Set times. 4. Set price and quota per class. 5. Save; system validates aircraft availability and time rules.
Postcondition	Flight is visible in search results

8.4 Use Case Diagram

Actors (Guest User, Registered User, Admin) and the system functions available to each. Book Flight includes Enter Passenger Details and Make Payment.

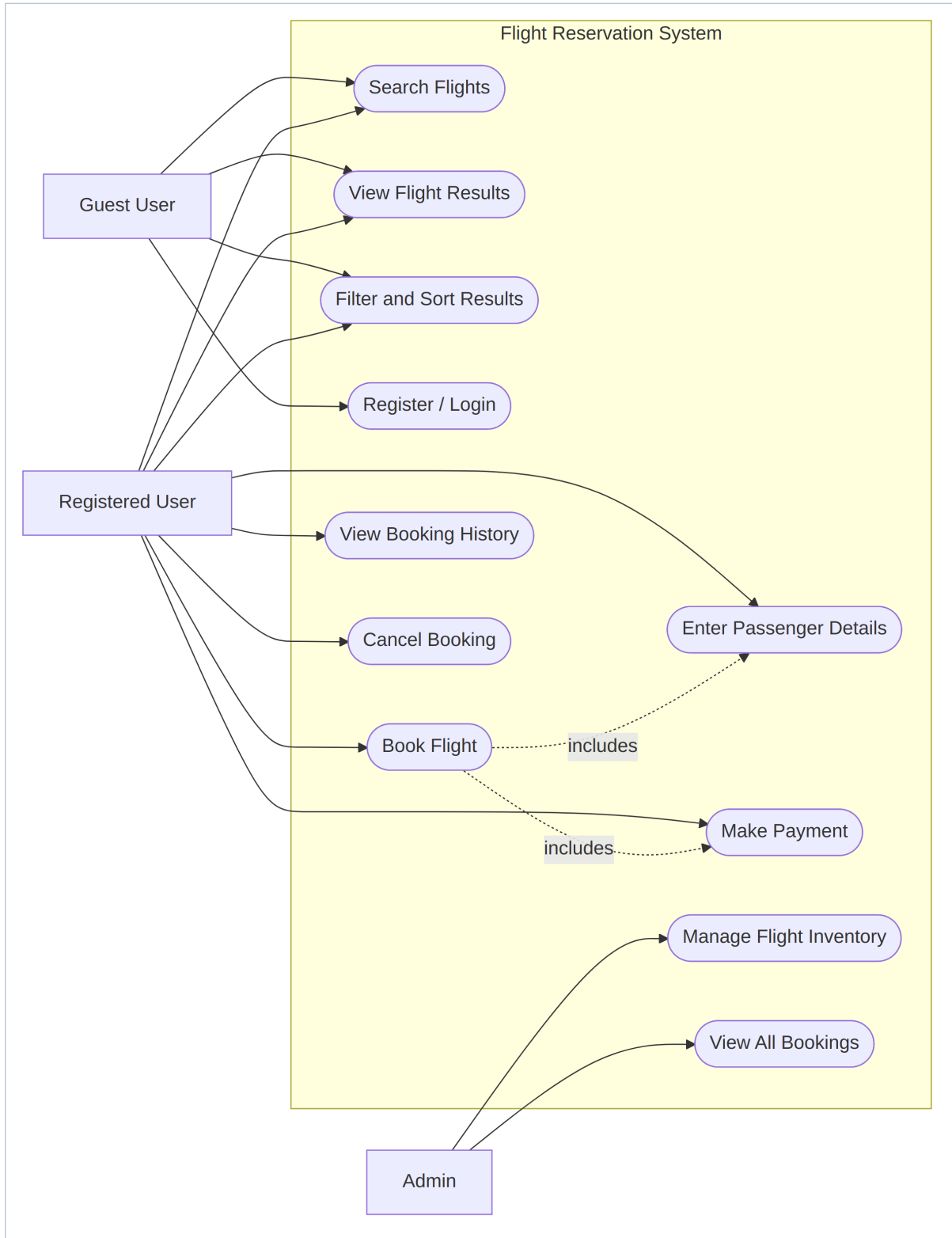


Figure 8.1 - Use case diagram

8.5 Activity Diagram: Booking Flow

Flowchart of the booking process from search to confirmation, including the login check and payment retry.

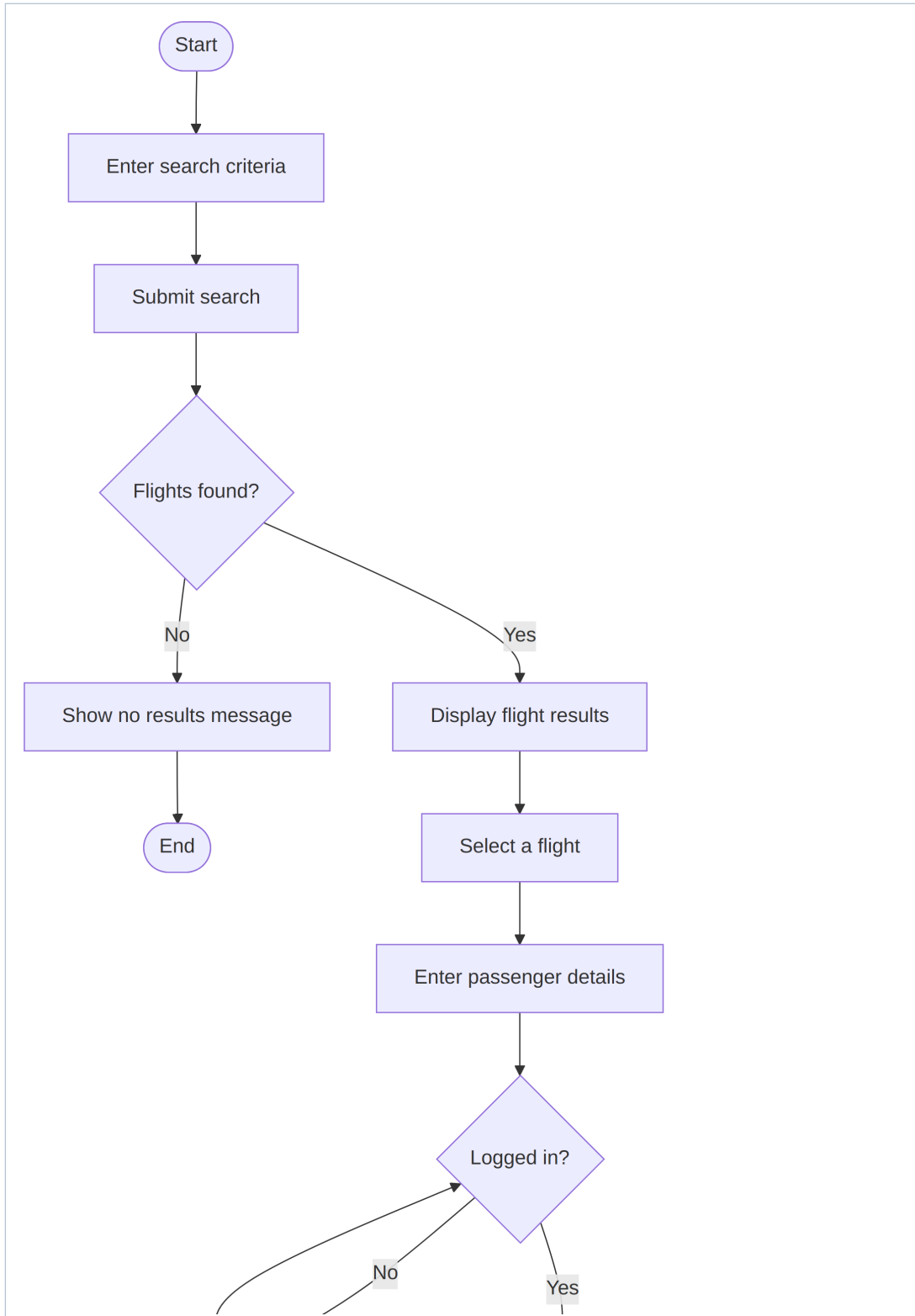


Figure 8.2 - Booking activity diagram (1 of 2)

8.5 Activity Diagram: Booking Flow (continued)

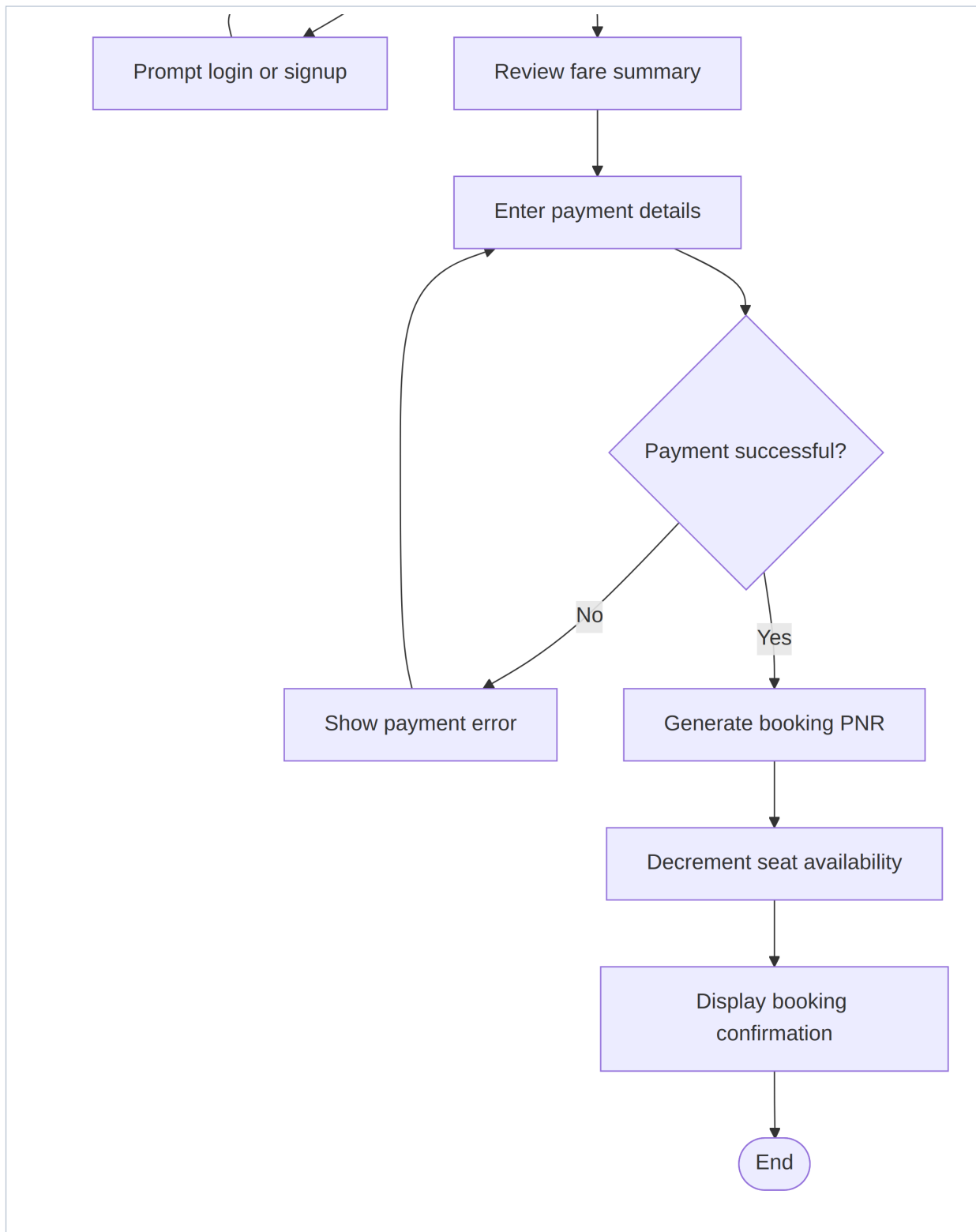


Figure 8.2 - Booking activity diagram (2 of 2)

8.6 Sequence Diagram

Message flow between the user, React front end, back-end API, database and mock payment service for flight search, booking and payment.

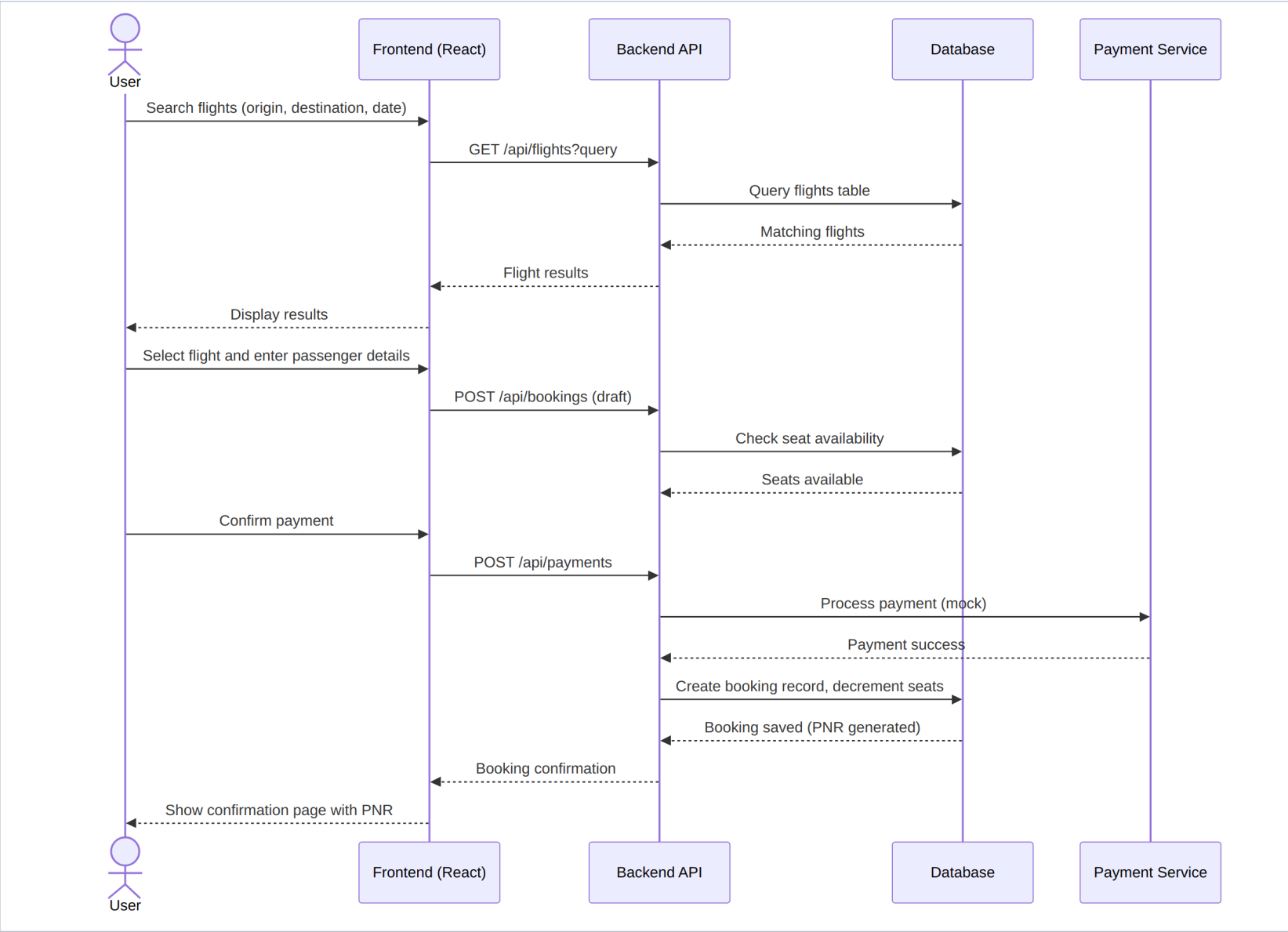


Figure 8.3 - Booking sequence diagram

8.7 Communication Diagram

Numbered message exchange (1 to 10) between the user, front end, Booking API, database and payment service when a booking is submitted.

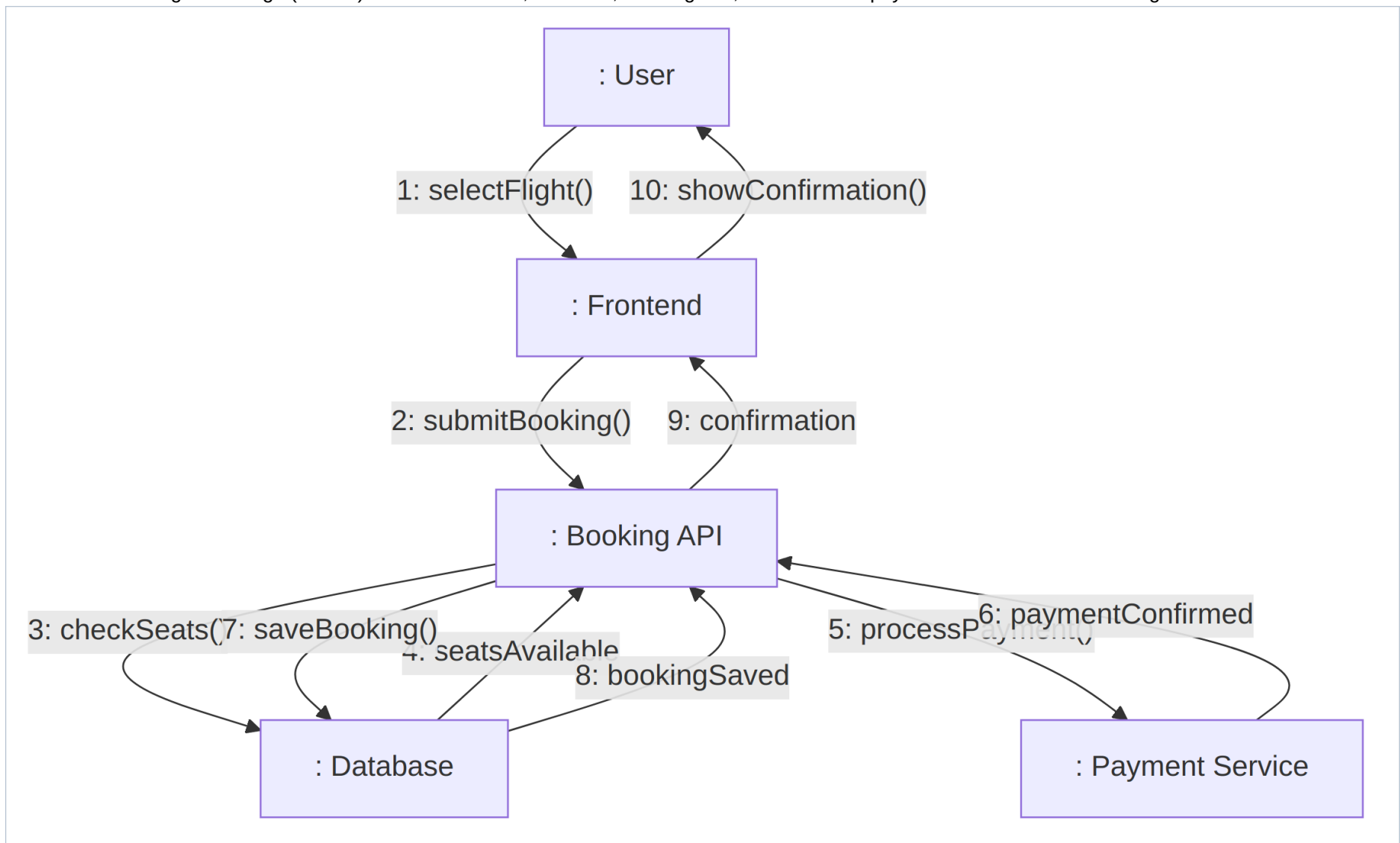


Figure 8.4 - Booking communication diagram

9. Suggested Build Phases

Phase	Deliverable	Requirements covered
1. Foundation	Database schema, airports CRUD, auth	FR-1, FR-2
2. Flights	Routes, aircraft, flight scheduling, search	FR-3, FR-4
3. Classes and Fares	Ticket classes, per-flight pricing	FR-5
4. Seating	Seat map builder, seat selection UI, holds	FR-6
5. Booking	Passenger details, simulated payment, PNR, e-ticket	FR-7, FR-8
6. Polish	Admin reports, emails, tests, deployment	FR-9, NFR

10. Appendix

10.1 Open Questions

- Which tech stack will be used (front end, back end, database)?
- Will the app be single-airline or multi-airline?
- Is round-trip pricing a sum of two legs or a separate fare?
- Are seat selection fees charged by class?

10.2 Traceability

Each functional requirement (FR-x.y) should map to at least one use case and one test case in the project test plan.